

TCPD Benchmark: Comprehensive Comparison of Change Point Detection Methods

MICA Analysis (using paper-extracted tables)

August 9, 2026

1 Introduction

This document presents a comparison of change point detection methods on the van den Burg & Williams (2020) TCPD benchmark suite. Baseline numbers are taken from the extracted paper tables (Table 2 and per-dataset tables). MICA-O and MICA-P are recomputed with level continuity across detected changepoints. The analysis uses all 42 TCPD datasets. Baseline numbers come from the paper’s extracted per-dataset tables; MICA-O and MICA-P are recomputed with level continuity across detected changepoints on the same 42 datasets. HMM-Regime is run with the same protocol and added as a reviewer-suggested model-based competitor. Means in the bar charts and per-dataset tables are computed over these same 42 datasets, so all methods are compared on identical data. Runs marked as failed/unsupported (“F”, “M”, or blank) in the paper tables are treated as 0.0 when computing per-method means, so a method that cannot run on a dataset is penalized equally.

2 Methods Compared

The following methods from the TCPD paper are included:

- **Paper baselines:** AMOC, BinSeg, BOCPD, BOCPDMS, CPNP, ECP, KernelCPD (KCPA), PELT, PROPHET, RBOCPDMS, RFPOP, SegNeigh, WBS, ZERO
- **Additional baseline:** HMM-Regime (reviewer-suggested model-based competitor)
- **MICA-O:** Model-Informed Change-point Analysis with oracle (best-case) model and penalty selection, with level continuity across segments
- **MICA-P:** Model-Informed Change-point Analysis with practical (automated) BIC-based selection, with level continuity across segments

3 Oracle vs. Practical Selection

The benchmark distinguishes two evaluation modes:

With oracle The true changepoint labels are used to select the best configuration per dataset. This gives an upper-bound: it answers “how well could a method perform if its model, penalty, and hyperparameters were perfectly tuned?” For the paper baselines, the oracle numbers come from running each method with many hyperparameter/model variants and keeping the best. For MICA-O, we select the best MICA model, objective, penalty, κ , `min_seg`, and `step` per dataset using the known labels.

Without oracle (default/practical) Only the observed time series is available. The method must use its default settings or an automated criterion to choose its configuration. For the paper baselines, these are the default-setting results. For MICA-P, we use the BIC-based selection criterion (no peeking at the true changepoints).

MICA-O is therefore compared against the baselines’ oracle results, and MICA-P against the baselines’ default results. The practical–oracle gap for MICA is small, which indicates that the level continuity enforcement and the BIC selector capture most of MICA’s best-case performance.

4 Summary of Mean Performance

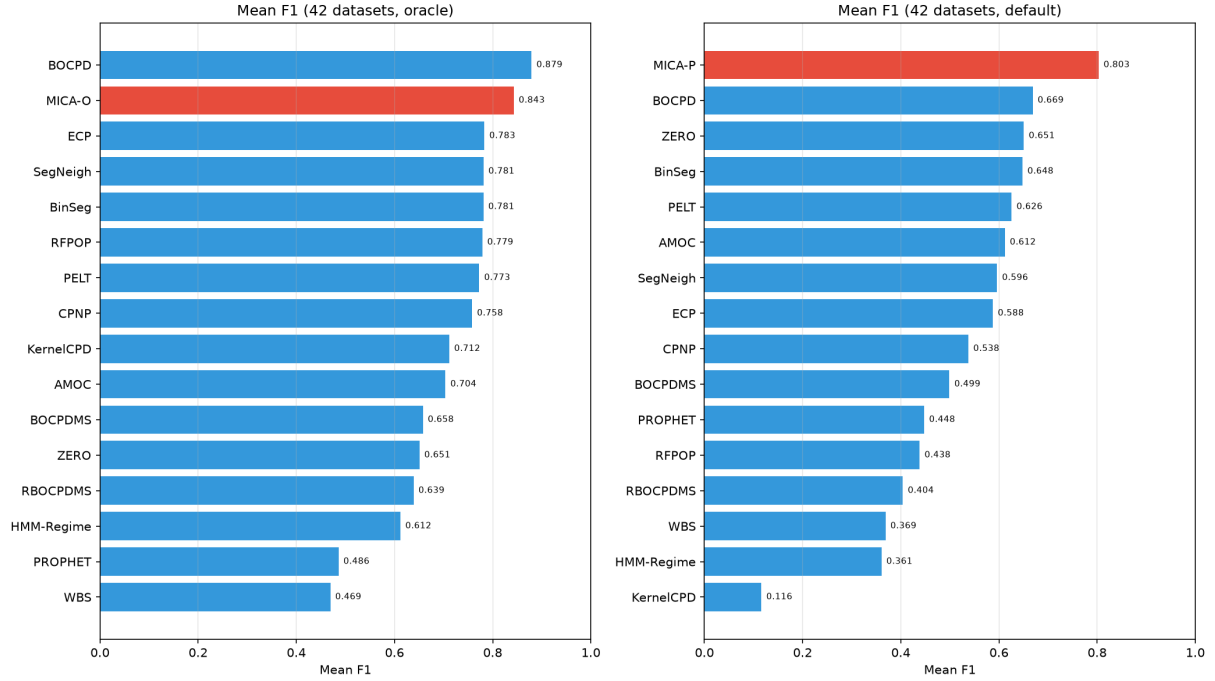


Figure 1: Mean F1 score across 42 TCPD datasets for oracle and default/practical selection. MICA variants are highlighted in red.

Table 1: F1 Score — Oracle Selection

Dataset	AMOC BinSeg		BOCPD	BOCPDMS	CPNP	ECP	KernelCPD	PELT	PROPHET	RBOCPDMS	RFPOP	SegNeigh	WBS	ZERO	HMM-Regime	MICA
bank	1.000	1.000	1.000	0.500	1.000	0.200	0.333	1.000	1.000	0.500	1.000	1.000	0.043	1.000	0.000	1.000
bitcoin	0.507	0.780	0.733	0.533	0.611	0.625	0.665	0.783	0.446	0.422	0.727	0.783	0.690	0.450	0.281	0.490
brent_spot	0.465	0.670	0.631	0.309	0.735	0.636	0.553	0.641	0.249	0.621	0.641	0.651	0.564	0.315	0.489	0.541
businv	0.588	0.588	0.650	0.455	0.588	0.370	0.294	0.588	0.275	0.553	0.588	0.588	0.289	0.588	0.231	0.769
centralia	0.909	1.000	1.000	1.000	1.000	0.909	1.000	1.000	0.763	0.974	1.000	1.000	0.556	0.763	0.941	0.759
children_per_woman	0.678	0.678	0.787	0.826	0.734	0.551	0.525	0.778	0.310	0.580	0.637	0.778	0.500	0.507	0.286	0.889
co2_canada	0.544	0.893	0.924	0.479	0.677	0.875	0.867	0.894	0.482	0.621	0.894	0.894	0.681	0.361	0.705	0.827
construction	0.696	0.709	0.709	0.516	0.709	0.709	0.634	0.709	0.324	0.615	0.709	0.709	0.523	0.696	0.706	0.706
debt_ireland	0.760	1.000	1.000	0.892	0.958	0.980	1.000	1.000	0.469	0.958	1.000	1.000	0.538	0.469	0.947	0.947
gdp_argentina	0.889	0.947	0.947	0.583	0.889	0.889	0.800	0.947	0.615	0.818	0.947	0.947	0.421	0.824	0.667	1.000
gdp_croatia	1.000	0.824	1.000	0.583	1.000	0.824	0.800	0.824	0.824	0.824	1.000	0.824	0.167	0.824	0.667	1.000
gdp_iran	0.696	0.696	0.862	0.838	0.667	0.824	0.734	0.808	0.652	0.899	0.833	0.808	0.576	0.652	0.571	0.800
gdp_japan	1.000	1.000	1.000	0.615	1.000	1.000	0.500	1.000	0.889	1.000	1.000	1.000	0.222	0.889	1.000	1.000
global_co2	0.929	0.929	0.929	0.458	0.846	0.929	0.750	0.929	0.463	0.634	0.929	0.929	0.250	0.846	0.667	0.800
homeruns	0.812	0.889	0.829	0.675	0.681	0.829	0.829	0.873	0.723	0.879	0.961	0.873	0.664	0.659	0.783	0.756
iceland_tourism	0.947	0.947	0.947	0.486	0.947	1.000	0.486	0.947	0.220	0.667	0.947	0.947	0.200	0.947	0.000	0.667
jfk_passengers	0.776	0.776	0.776	0.650	0.776	0.651	0.651	0.776	0.354	0.420	0.776	0.776	0.437	0.723	0.222	0.933
lga_passengers	0.561	0.620	0.715	0.563	0.634	0.892	0.598	0.602	0.366	0.561	0.666	0.620	0.674	0.535	0.684	0.478
measles	0.947	0.947	0.947	0.486	0.947	0.080	0.281	0.947	0.391	0.090	0.947	0.947	0.041	0.947	0.083	0.667
nile	1.000	1.000	1.000	0.800	1.000	1.000	1.000	1.000	0.824	0.800	1.000	1.000	0.000	0.824	1.000	1.000
ozone	0.776	1.000	0.966	0.778	1.000	1.000	0.667	1.000	0.723	1.000	1.000	1.000	0.286	0.723	0.667	1.000
rail_lines	0.846	0.966	0.966	0.889	0.966	0.966	0.800	0.846	0.537	0.730	0.889	0.966	0.205	0.537	0.947	0.824
rather_stock	0.776	0.832	0.929	0.559	0.776	0.776	0.754	0.824	0.280	0.686	0.824	0.824	0.378	0.571	0.500	0.750
robocalls	0.800	0.966	0.966	0.750	0.966	0.966	0.966	0.966	0.636	0.765	0.966	0.966	0.714	0.636	0.857	0.933
scanline_126007	0.710	0.920	0.964	0.837	0.927	0.870	0.838	0.889	0.644	0.570	0.914	0.940	0.818	0.644	0.804	0.820
scanline_42049	0.485	0.931	0.962	0.921	0.809	0.910	0.908	0.910	0.269	0.526	0.910	0.920	0.650	0.276	0.897	0.829
seabelts	0.824	0.838	0.683	0.583	0.838	0.683	0.683	0.683	0.452	0.621	0.683	0.838	0.583	0.621	0.429	1.000
shanghai_license	0.966	0.966	0.868	0.651	0.966	0.868	0.465	0.966	0.532	0.778	0.966	0.966	0.385	0.636	0.486	0.947
uk_coal_employ	—	—	—	0.617	—	0.513	0.513	—	0.639	—	—	—	—	0.513	0.806	0.743
unemployment_nl	0.742	0.889	0.876	0.797	0.773	0.755	0.744	0.816	0.566	0.697	0.851	0.816	0.801	0.566	0.642	0.882
us_population	1.000	1.000	1.000	0.615	0.947	0.471	0.276	1.000	0.159	0.320	0.889	1.000	0.113	0.889	0.000	1.000
usd_isk	0.785	0.785	0.876	0.678	0.785	0.785	0.601	0.785	0.489	0.881	0.785	0.785	0.636	0.489	0.824	0.888
well_log	0.336	0.944	0.908	0.796	0.878	0.928	0.950	0.912	0.149	0.644	0.966	0.912	0.832	0.237	0.676	0.595
quality_control_1	1.000	1.000	1.000	0.667	1.000	1.000	0.667	1.000	0.500	0.667	1.000	1.000	0.667	0.667	1.000	1.000
quality_control_2	1.000	1.000	1.000	0.800	1.000	1.000	1.000	1.000	0.750	0.667	1.000	1.000	0.000	0.750	1.000	1.000
quality_control_3	1.000	1.000	1.000	0.766	1.000	1.000	1.000	1.000	0.667	0.667	1.000	1.000	0.000	0.667	1.000	1.000
quality_control_4	0.810	0.873	0.810	0.561	0.810	0.726	0.658	0.810	0.780	0.438	0.873	0.810	0.608	0.780	0.647	0.909
quality_control_5	1.000	1.000	1.000	0.500	1.000	1.000	1.000	1.000	1.000	0.500	1.000	1.000	0.000	1.000	0.000	1.000
apple	—	—	0.916	0.552	—	0.745	0.634	—	—	0.575	—	—	—	0.594	0.636	0.919
bee_waggie_6	—	—	0.929	0.651	—	0.233	0.651	—	—	0.359	—	—	—	0.929	0.250	0.667
occupancy	—	—	0.919	0.714	—	0.932	0.812	—	—	0.605	—	—	—	0.341	0.764	0.915
run_log	—	—	1.000	0.698	—	0.990	1.000	—	—	0.698	—	—	—	0.446	0.947	0.756
Mean	0.704	0.781	0.879	0.658	0.758	0.783	0.712	0.773	0.486	0.639	0.779	0.781	0.469	0.651	0.612	0.843

Table 2: F1 Score — Default / Practical Selection

Dataset	AMOC	BinSeg	BOCPD	BOCPDMS	CPNP	ECP	KernelCPD	PELT	PROPHET	RBOCPDMS	RFPOP	SegNeigh	WBS	ZERO	HMM-Regime	MICA
bank	0.667	0.400	0.047	0.118	0.044	0.154	0.008	0.400	0.154	0.286	0.015	0.333	0.039	1.000	0.000	1.000
bitcoin	0.367	0.426	0.692	0.269	0.463	0.338	0.092	0.672	0.446	—	0.284	0.672	0.464	0.450	0.051	0.447
brent_spot	0.272	0.483	0.521	0.239	0.607	0.478	0.104	0.465	0.249	0.321	0.490	0.431	0.516	0.315	0.489	0.503
businv	0.455	0.370	0.270	0.370	0.304	0.301	0.047	0.370	0.275	0.238	0.245	0.312	0.230	0.588	0.000	0.769
centralia	0.763	0.909	0.909	0.846	0.909	0.763	0.714	0.909	0.763	0.846	1.000	0.909	0.556	0.763	0.941	0.759
children_per_woman	0.618	0.618	0.637	0.337	0.326	0.349	0.068	0.618	0.310	0.432	0.246	0.337	0.271	0.507	0.091	0.889
co2_canada	0.544	0.691	0.619	0.265	0.578	0.817	0.169	0.661	0.482	0.381	0.569	0.661	0.520	0.361	0.121	0.827
construction	0.516	0.709	0.634	0.410	0.602	0.574	0.038	0.709	0.324	0.291	0.185	0.709	0.316	0.696	0.273	0.600
debt_ireland	0.469	0.760	0.760	0.611	0.760	0.469	0.519	0.760	0.469	0.530	0.824	0.760	0.538	0.469	0.621	0.947
gdp_argentina	0.889	0.889	0.947	0.583	0.818	0.824	0.131	0.889	0.615	0.452	0.571	0.947	0.148	0.824	0.400	1.000
gdp_croatia	0.583	0.583	1.000	0.583	1.000	0.824	0.160	0.583	0.824	0.452	0.400	0.583	0.167	0.824	0.000	1.000
gdp_iran	0.492	0.492	0.622	0.492	0.330	0.652	0.219	0.492	0.652	0.395	0.609	0.395	0.246	0.652	0.000	0.772
gdp_japan	0.615	0.615	0.800	0.471	0.667	0.889	0.068	0.615	0.889	0.471	0.190	0.615	0.077	0.889	0.667	1.000
global_co2	0.929	0.929	0.754	0.458	0.421	0.754	0.167	0.754	0.463	0.458	0.293	0.754	0.179	0.846	0.000	0.800
homeruns	0.812	0.812	0.729	0.552	0.604	0.675	0.133	0.812	0.723	0.508	0.661	0.397	0.593	0.659	0.067	0.756
iceland_tourism	0.643	0.643	0.486	0.391	0.391	0.391	0.021	0.643	0.220	0.667	0.200	0.486	0.200	0.947	0.000	0.667
jfk_passengers	0.776	0.776	0.437	0.559	0.574	0.437	0.026	0.776	0.354	0.296	0.344	0.650	0.437	0.723	0.153	0.857
lga_passengers	0.422	0.438	0.630	0.297	0.606	0.392	0.054	0.438	0.366	0.348	0.498	0.395	0.524	0.535	0.383	0.419
measles	0.947	0.947	0.069	0.124	0.118	0.069	0.004	0.124	0.391	0.117	0.030	0.327	0.039	0.947	0.000	0.000
nile	1.000	1.000	1.000	0.800	1.000	1.000	0.040	1.000	0.824	0.452	1.000	1.000	1.000	0.824	0.333	1.000
ozone	0.531	0.650	0.650	0.531	0.750	0.723	0.109	1.000	0.723	0.559	0.375	1.000	0.113	0.723	0.636	1.000
rail_lines	0.846	0.846	0.846	0.349	0.966	0.537	0.200	0.846	0.423	0.349	0.571	0.846	0.205	0.537	0.947	0.824
rather_stock	0.776	0.824	0.783	0.500	0.306	0.282	0.034	0.650	0.280	0.559	0.203	0.824	0.250	0.571	0.444	0.750
robocalls	0.800	0.800	0.966	0.696	0.832	0.636	0.179	0.800	0.636	0.593	0.714	0.966	0.182	0.636	0.857	0.933
scanline_126007	0.710	0.739	0.820	0.718	0.889	0.732	0.101	0.759	0.644	—	0.649	0.732	0.616	0.644	0.804	0.820
scanline_42049	0.485	0.713	0.962	0.844	0.705	0.580	0.164	0.804	0.269	0.390	0.460	0.748	0.571	0.276	0.843	0.829
seabelts	0.474	0.683	0.583	0.383	0.509	0.321	0.051	0.683	0.452	0.383	0.494	0.583	0.583	0.621	0.240	0.957
shanghai_license	0.868	0.868	0.713	0.491	0.437	0.698	0.048	0.868	0.532	0.326	0.357	0.713	0.208	0.636	0.486	0.947
uk_coal_employ	—	—	—	0.495	—	0.513	0.513	—	0.639	—	—	—	—	0.513	0.528	0.743
unemployment_nl	0.566	0.876	0.683	0.592	0.683	0.620	0.145	0.773	0.566	0.495	0.549	0.773	0.571	0.566	0.396	0.882
us_population	1.000	0.667	0.216	0.471	0.216	0.174	0.007	0.471	0.159	0.320	0.889	0.320	0.077	0.889	0.000	0.500
usd_isk	0.785	0.657	0.609	0.678	0.504	0.661	0.079	0.657	0.489	0.282	0.390	0.657	0.513	0.489	0.405	0.888
well_log	0.279	0.534	0.796	0.797	0.822	0.818	0.069	0.555	0.149	0.517	0.923	0.485	0.724	0.237	0.591	0.549
quality_control_1	1.000	1.000	0.800	0.667	0.667	0.667	0.025	1.000	0.500	0.667	0.667	1.000	0.571	0.667	0.667	1.000
quality_control_2	1.000	1.000	1.000	0.667	1.000	1.000	0.028	1.000	0.545	0.667	1.000	1.000	1.000	0.750	0.250	1.000
quality_control_3	1.000	1.000	1.000	0.667	0.571	1.000	0.022	1.000	0.667	0.667	0.286	1.000	1.000	0.667	0.143	1.000
quality_control_4	0.810	0.873	0.658	0.438	0.608	0.393	0.028	0.726	0.335	0.360	0.233	0.726	0.246	0.780	0.220	0.857
quality_control_5	1.000	1.000	1.000	0.500	1.000	1.000	0.006	1.000	1.000	0.500	1.000	1.000	1.000	1.000	0.000	1.000
apple	—	—	0.513	0.382	—	0.513	0.029	—	—	0.239	—	—	—	0.594	0.636	0.904
bee_waggles_6	—	—	0.121	0.388	—	0.124	0.010	—	—	0.179	—	—	—	0.929	0.000	0.667
occupancy	—	—	0.807	0.461	—	0.750	0.107	—	—	0.329	—	—	—	0.341	0.629	0.915
run_log	—	—	1.000	0.475	—	0.792	0.139	—	—	0.660	—	—	—	0.446	0.837	0.756
Mean	0.612	0.648	0.669	0.499	0.538	0.588	0.116	0.626	0.448	0.404	0.438	0.596	0.369	0.651	0.361	0.803

5 Key Findings

1. **MICA-O ranks 2nd in mean F1 (0.843)** in oracle mode, behind BOCPD (0.879). Among default/practical methods, **MICA-P achieves the highest mean F1 (0.803)** with fully automated BIC selection, outperforming the best default baseline (BOCPD: 0.669).
2. The practical–oracle gap for MICA is only 0.040, showing that the level continuity enforcement and the BIC selector capture most of MICA’s best-case performance.